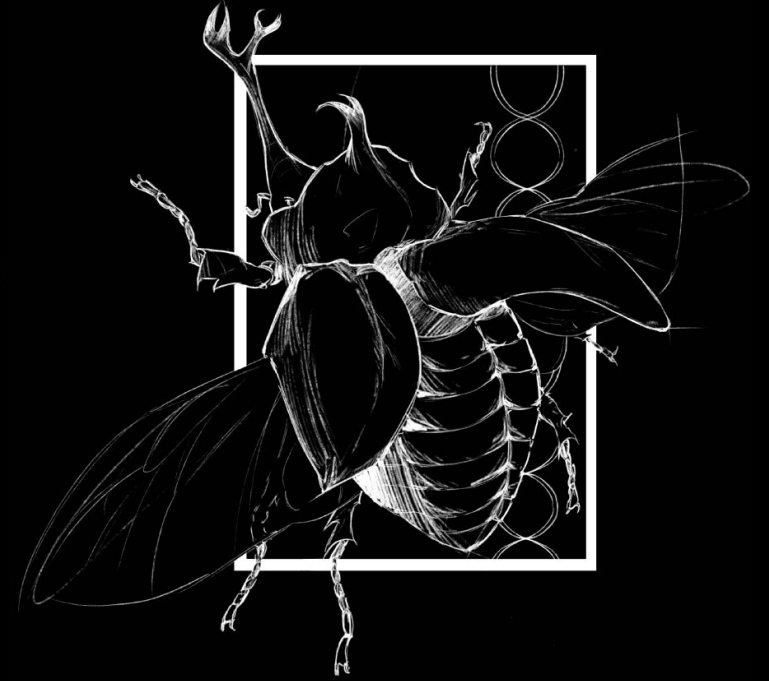




Background

Caffeine is a purine alkaloid that plants use as a chemical defense against herbivores and pathogens. By interfering with insect nervous system function, caffeine deters feeding and improves plant survival and reproductive success. Despite this apparent advantage, caffeine production is far from universal across the tribe Coffeae (Rubiaceae). While some species produce substantial amounts of caffeine, others produce entirely different alkaloids or none at all. This uneven distribution raises a compelling evolutionary question: did caffeine biosynthesis arise once in a common ancestor and persist through inheritance, or has it appeared and disappeared multiple times across the evolutionary history of the group?

To investigate this, we tested three competing explanations:

- **Hypothesis 1: Caffeine biosynthesis has been gained and lost multiple times across Coffeae evolutionary history**
- **Hypothesis 2: Closely related species are more likely to share caffeine production patterns because of shared ancestry**
- **Alternative Hypothesis: Caffeine evolved once in an early ancestor and was conserved across the group due to its functional importance**

Study Design & AI

This study integrated published phytochemical data with phylogenetic comparative methods to trace the evolutionary history of caffeine biosynthesis across Coffeae.

Data Sources

- Caffeine concentration data (expressed as percentage dry matter basis) were compiled from published phytochemistry literature covering 143 Coffeae species, of which 50 had quantified caffeine values
- A phylogenetic tree was retrieved from Open Tree of Life using the rotl package in R, with 38 species successfully matched to the tree for downstream analysis

Analytical Pipeline

- Caffeine concentrations were converted to binary presence/absence data (0 = absent, 1 = present) for discrete trait modeling
- Two Markov models of trait evolution were fitted and compared to identify the best-fit model for caffeine transitions
- Blomberg's K statistic was calculated to quantify the degree of phylogenetic signal in caffeine concentration across species
- Ancestral state reconstruction and stochastic character mapping were performed using the phyttools package, with 1000 simulations to estimate the average number of gains and losses across the phylogeny

AI Assistance (Claude, Anthropic)

- R scripts were generated with AI assistance for data extraction, figure creation, and phylogenetic analysis
- AI guided the initial framing of research questions and helped identify relevant literature
- Species name matching and phylogenetic tree retrieval were completed with AI support
- Statistical results were interpreted collaboratively with AI assistance



Model Fitting

To determine whether caffeine is gained and lost at equal or different rates across the Coffeae phylogeny, two Markov models of discrete trait evolution were compared using the Akaike Information Criterion (AIC), where lower values indicate a better fit.

- Equal Rates model (ER): both gains and losses occur at the same rate — AIC = 39.18 (best fit)
- All Rates Different model (ARD): gains and losses are allowed to occur at different rates — AIC = 39.32

The simpler Equal Rates model provided the better fit, suggesting there is no strong statistical support for asymmetric rates of caffeine gain versus loss at the macroevolutionary scale. The ER transition rate was estimated at 0.058, representing the overall rate of state change across the phylogeny.



Discussion

Taken together, these findings support a dynamic and complex evolutionary history of caffeine biosynthesis in Coffeae rather than a single origin with strict conservation.

Stochastic character mapping revealed an average of 2.66 independent gains and 9.39 losses of caffeine across 1000 simulated evolutionary histories, providing clear support for Hypothesis 1. The ancestral state reconstruction further showed that caffeine was most likely present in the common ancestor of most sampled species, with the 7 caffeine-free species representing comparatively recent losses rather than an ancient absence. This partially supports the Alternative Hypothesis as well, pointing to a picture where caffeine arose early and was largely maintained, but with repeated losses in specific lineages over time.

The significant phylogenetic signal in caffeine concentration (K = 0.89, p = 0.017) supports Hypothesis 2, confirming that shared ancestry meaningfully predicts caffeine levels across species. It is worth noting that this result only reached statistical significance with the expanded 38-species dataset, underscoring how sensitive phylogenetic comparative analyses are to sample size.

The preference for the Equal Rates model over the asymmetric ARD model was somewhat unexpected given the biological complexity of the caffeine biosynthesis pathway. Future work with a larger species set and a species-specific Coffeae phylogeny may reveal more nuanced asymmetries in transition rates. The synthetic tree from Open Tree of Life, while useful for exploratory analysis, may not fully capture the fine-scale relationships within the tribe.

Results

Caffeine Distribution Across Species

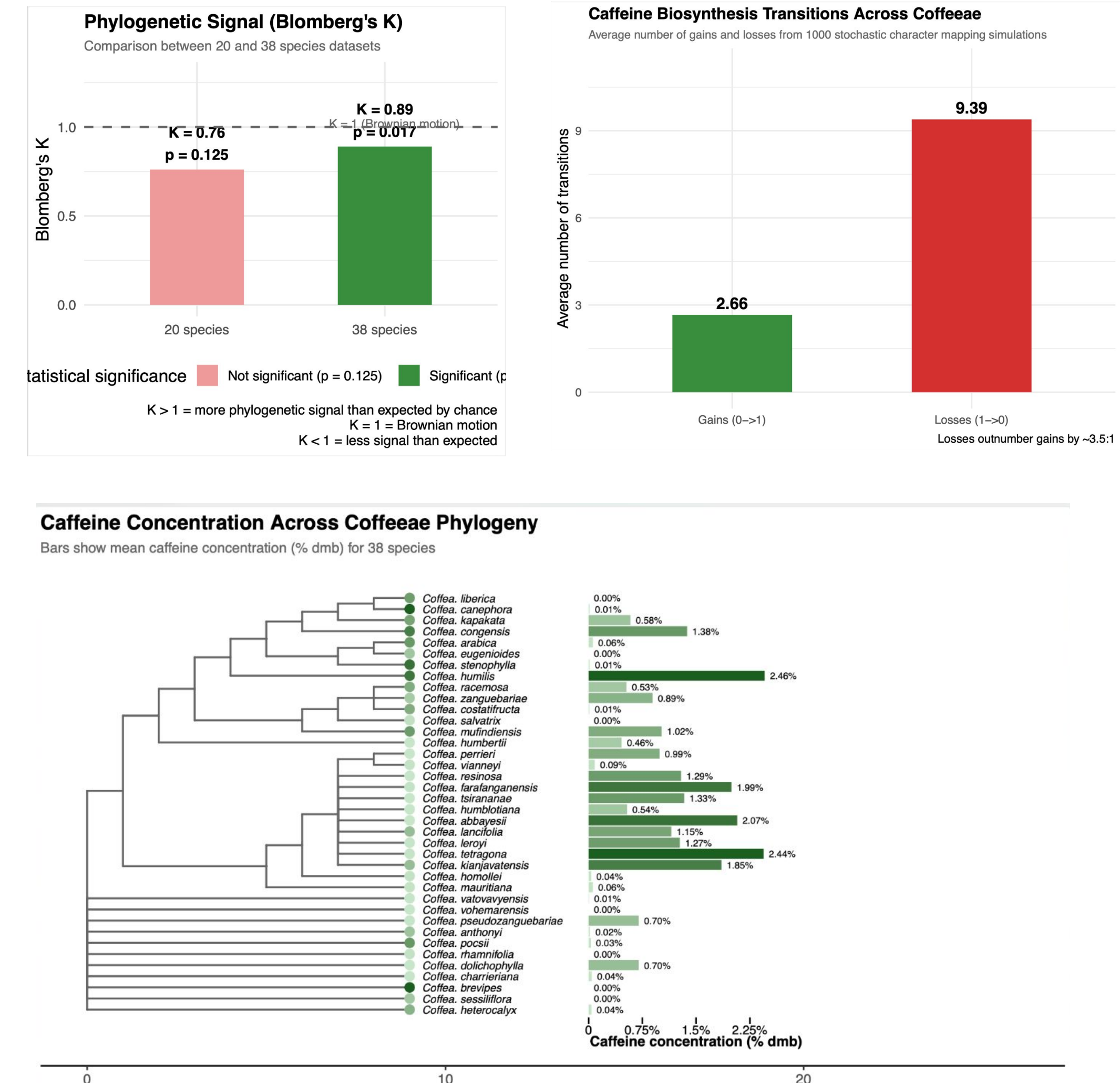
- Of the 38 Coffeae species included in this analysis, 31 (82%) had detectable caffeine and 7 (18%) were caffeine-free
- Caffeine concentrations spanned a wide range, from 0.00% to 2.46% dry matter basis, with a mean of 0.63%
- Coffea brevipes showed the highest caffeine concentration at 2.46%, while caffeine-free species were predominantly rare endemic species from Madagascar

Phylogenetic Signal in Caffeine Concentration

- Blomberg's K = 0.89, p = 0.017, indicating statistically significant phylogenetic signal in caffeine concentration
- This means that closely related species tend to have more similar caffeine levels than would be expected by chance
- This result provides strong support for Hypothesis 2 and suggests that evolutionary history plays a meaningful role in shaping caffeine production across species

Rates of Evolutionary Gain and Loss

- Stochastic character mapping across 1000 simulations estimated an average of 2.66 gains and 9.39 losses of caffeine biosynthesis across the phylogeny
- Caffeine was lost approximately 3.5 times more frequently than it was gained
- This pattern suggests that while caffeine provides a defensive advantage, the ability to produce it is repeatedly and independently lost across lineages



Conclusions & Acknowledgements

Conclusions

- Caffeine biosynthesis has been gained approximately 3 times and lost approximately 9 times across the Coffeae phylogeny, strongly supporting Hypothesis 1
- Closely related species share significantly similar caffeine concentrations (K = 0.89, p = 0.017), supporting Hypothesis 2
- The common ancestor of most sampled species very likely produced caffeine, with independent losses occurring in at least 7 lineages
- The Equal Rates model best describes the data, suggesting caffeine gains and losses occur at similar rates
- The frequency of loss relative to gain (~3.5:1) suggests that maintaining caffeine biosynthesis requires ongoing selective pressure

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